Besher Kitaz

Software Engineer
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EDUCATION

Berea College **Sep 2024 - May 2028**

Computer Science

- **Achievements:** Berea College Full Ride Scholarship, valued at $204,000, Dean's List: Fall 2024 and Fall 2025
- **Coursework:** Software Design and Implementation, Data Structures, Scalable Algorithms and Objects, Theory of Computation, Database Systems, Operating Systems and Virtual Machines

EXPERIENCE

Purdue University College of Engineering | *Software Engineer* **Jun 2026 - Aug 2026**

- Led the design and development of SafetyHub, a full-stack makerspace certification management system using React, TypeScript, Express.js, PostgreSQL, and Axios.
- Designed and implemented a multi-level certification system using a Directed Acyclic Graph (DAG) to model training prerequisites, validate certification eligibility, and track certifications issued by authorized staff and mentors.
- Engineered a role-based access control (RBAC) system to manage safety training and tool certifications for a potential user base of 7,000+ Purdue students and staff, with an estimated 120 administrative hours saved annually.
- Implemented student affiliation tracking and data collection across 50+ schools and departments, enabling institutional analytics, usage reporting, and data-driven funding requests.
- Containerized the application using Docker, ensuring consistent development environments and streamlined deployment.
- Built administrative dashboards with searchable certification records, enabling staff to track training progress, verify tool eligibility, and manage certification status.

Berea College | *Software Engineer Intern* **Jun 2025 - Aug 2025**

- Collaborated with 18 engineers to maintain and extend two production open-source systems used across Berea College, delivering bug fixes, new features, and automated tests.
- Refactored and improved the Employment Status Form system used by 1,700+ students and 23 administrators, streamlining workflows and improving maintainability
- Implemented role-based data visibility, updating database schemas, UI components, and backend logic for secure and efficient information access.
- Contributed to 17+ merged Pull Requests, with 76+ commits, participated in code reviews, and feature development, shipping AJAX-driven endpoints, and database schema upgrades for the CELTS (Center for Excellence in Learning Through Service), Berea's community engagement hub website.
- Managed GitHub repositories for CELTS and LSF (https://github.com/BCStudentSoftwareDevTeam), ensuring an organized codebase and easy access for the development team

Part-time Mentor and Web Developer **Sep 2024 - Jan 2026**

- Developed a responsive website for the team with modern technologies, promoting the team with 4 major sponsors.
- Mentored 18 high-school students in Java and robotics software (e.g., WPILib), guiding them through the robot design and build process for the FIRST Robotics Competition, resulting in a functional robot that qualified for regional events
- Published the project on GitHub (https://github.com/Rat-Fight/rat-fight.github.io), enabling public access to the codebase and facilitating collaboration and feedback from the developer community

Paper Airplanes | *Volunteered Python and Machine Learning Instructor* **Oct 2023 - Nov 2024**

- Coached three semesters of Women in Tech cohorts from underrepresented communities, teaching fundamental Python concepts with Flask and Peewee, and guided participants to complete capstone projects that demonstrated their new skills
- Developed a curriculum plan with a team of 6 mentors, which included topics on basic programming, machine learning, and data analysis, empowering over 20 students to land jobs.

SKILLS

- **Languages and Technologies**: Python, JavaScript, TypeScript, C++, MySQL, HTML, CSS, MySQL, PostgreSQL, Prisma, Peewee
- **Frameworks**: React, React Navigation, Express.js, Flask, Bootstrap, Material UI, NumPy, Pandas, Axios
- **Tools and Practices**: Linux Command Line, Git, GitHub, Open-Source Development, Docker and Dev Ops
- **Non-Technical**: Agile, Peer Review, Cross-team Collaboration, Problem Solving, Adaptability, Teamwork

PROJECTS

Berea Bazaar | GitHub repository

- Built and deployed a Berea-exclusive marketplace in 24 hours with college-authenticated access, secure data handling, and student-focused buying and selling features.
- Accelerated development with V0 by Vercel and iterative prompt engineering, refining AI-generated components into a fully functional platform

Full-Stack AI Flash Card App | GitHub repository | ***Personal Project***

- Built a full-stack flashcard web app using React, Flask, and MySQL, implementing a spaced-repetition algorithm and integrating the Google Gemini API to auto-generate study cards.